

Kishoreganj University
2nd Year 2nd Semester B. Sc. (Engg.) Final Examination-2023
Department of Computer Science and Engineering
CSE 2207: Database (3 Credits)

Time: 3 Hours

Full Marks: 70

Figures shown in the right margin indicate full marks.
Answer 05 out of 07 questions.

- 1
 - a. What is a Database Management System (DBMS)? Explain its importance. 4
 - b. How do you convert an ER diagram into a relational model? Provide a brief explanation. 5
 - c. What is database engine? Describe the components of database engine with necessary diagrams. 5
- 2
 - a. Explain the difference between primary key, foreign key, and candidate key with examples. 4
 - b. Suppose a hospital management system needs to maintain records of **patients, doctors, rooms, and appointments**. Each patient has a unique patient ID, name, age, gender, contact details, and medical history, while each doctor has a doctor ID, name, specialization, department, and contact details. A patient can book multiple appointments, each assigned to one doctor, but a doctor can have multiple appointments, with details such as date, time, and status (confirmed, canceled, or completed). Patients may be admitted to hospital rooms, where each room has a room number, type (ICU, General, Private), and availability status. 10

Draw an ER diagram representing these entities, attributes, and relationships, ensuring proper cardinalities and primary keys.

- 3
 - a. Define functional dependency and explain its importance in database design. 3
 - b. What is the difference between BCNF and 3NF? Provide an example to illustrate the difference. 3
 - c. Step by step convert the following table into 1NF, 2NF, and 3NF with appropriate details. 8

StdID	StdName	Course	InstID	InstName	InstPhone
101	Alice	Data Structures, DBMS	101, 102	Dr. Smith, Dr. Lee	1234, 5678
102	Bob	DLD, OS	101, 103	Dr. Smith, Dr. Brown	1234, 9876

- 4
 - a. What is a JOIN operation? What are stored procedures, and how do they extend DBMS functionality? 4
 - b. What is View? Why view is needed in database system? 2
 - c. Consider the following relations and answer the following queries using SQL: 8

Patients (PatientID, Name, Age, Gender, Address)

Doctors (DoctorID, Name, Specialty, Department)

Appointments (AppointmentID, PatientID, DoctorID, Date, Time, Status)

Treatments (TreatmentID, PatientID, DoctorID, Diagnosis, Medication)

Departments (DepartmentID, DepartmentName)

- i) Retrieve the names of all patients who have an appointment scheduled.
- ii) Find the names of patients who have been diagnosed with "Diabetes".
- iii) Get the Number of Appointments Scheduled by Each Patient
- iv) Find patients who have appointments with both a "Cardiologist" and a "Neurologist".
- v) Retrieve the names of patients who have never scheduled an appointment.
- vi) Find the names of patients who have appointments with the same doctor on different dates.

- 5
- a. Explain the different types of storage in DBMS, including RAID levels. 5
 - b. What is semi-structured data? Give a comparison between XML data and JSON data. 4
 - c. Discuss the ACID properties required for maintaining data integrity. 5
- 6
- a. What are the purposes to place an index in a database? Give a brief classification of index structures? 4
 - b. What is the difference between hashing and indexing? Why do you need multilevel indexing? 4
 - c. Suppose a B⁺ Tree of order 3 stores values 10, 20, 30, 40, 50, 60, 70, 80, 90, 100. 6
 - (i) Draw the B⁺ Tree after all insertions.
 - (ii) Delete 40 and update the tree.
- 7
- a. What are the main types of database architectures? Draw the centralized architecture of database system and describe different types of database users. 10
 - b. What is the difference between DDL (Data Definition Language) and DML (Data Manipulation Language)? Give examples. 4